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OOPs LAB

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ROLL NO: 20BCS021

SUBJECT CODE: CEN 591

SEMESTER: 5th

COURSE: B.TECH.(COMPUTER ENGG.)

DEPT: DEPT OF COMPUTER ENGG.

SUBMITTED TO:

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S. NO.	DATE	PROGRAM	PAGE	SIGN
1		Write a program to implement STUDENT class consisting of name, enrollId and marks as class data members. Create three objects for the class using the concept of array of objects. Write member functions to read and display the student information. Also write the main program to create objects and call the member functions from the class.	4	
2		Write a C++ program handling the following details for students and staff using inheritance. • Student Details: name, address, percentage, marks • Staff Details: name, address, salary Create appropriate base and derived classes. Input the details and output them.	10	
3		Write a C++ program to perform the addition of two time objects in hour and minute format, display the result in hour:minute format using object as a function argument.	13	
4		Write a C++ program based on following scenario: Consider an example of a bookshop which sells books and video tapes. These two classes are inherited from the base class called media. The media class has command data members such as title and publication. The book class has data members for storing a number of pages in a book, and the tape class has the playing time in a tape. Each class will have member functions such as read() and show(). In the base class, these members have to be defined as virtual functions. Write a program which models the class hierarchy for the bookshop and processes objects of these classes using pointers to the base class.	15	
5		Write a C++ program to overload [] operator for the following scenario: Create a class AccountBook that contains account holder details such as name and account number. Take input for 5 account holders in the account table. When we enter account	17	

		number, then the program prints account holder name while entering of account holder name, it prints account number of holder.		
6		Write a C++ Program to implement Complex class representing complex numbers. Include operator functions to overload the operators +=, -=, *=, /= and the << operator for the class. Here << operator should be used for printing the results of complex number operation.	19	
7		Design classes such that they support the following statements: Rupee rl, r2; Dollar dl, d2; dl = r2; // converts rupee (Indian currency) to dollar (US currency) r2 = d2; // converts dollar (US currency) to rupee (Indian currency) Write a complete program which does such conversions according to the world market value.	21	
8		Write suitable C++ program to implement following OOPS concepts: (a) Pure Virtual Function (b) Pointers to Derived Class Object (c) Virtual Destructor (d) Overloading through friend function	23	
9		Write a java program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangleand Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea( ) that prints the area of the given shape.	25	
10		Write a java program that implements a multi thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.	27	

FAIZAN CHOUDHARY

20BCS021

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CODE: (code pasted in this format for readability)

```
#include <iostream>
#include <string>
using namespace std;

class Student {
    int id;
    //double attendance;
public:
    string name;
    int marks[5];
    // int scholarship;

    void getData();
    void showData(bool flag);
    int getId() {
        return id;
    }
    void setId(int stu_id) {
        id = stu_id;
    }
};

const int MAX = 100;
Student s[MAX];

void Student :: getData() {
    cout<<"\nEnter details for student:\n";
    cout<<"\nEnter student id: ";
    int id;
    cin>>id;
    setId(id);
    cout<<"\nEnter name of the student: ";
    cin.get();
    getline(cin, name);
    cout<<"\nEnter marks in 5 subjects (out of 100): ";
    for (int i=0; i<5; i++) {
        int mk;
        A:
        cin>>mk;
        if (mk > 100 || mk < 0) {
            cout<<"\nInvalid marks! Enter again...";
            goto A;
        }
        else
```

```
            marks[i] = mk;
    }
}

void Student :: showData(bool flag) {
    if (flag) {
        cout<<"|\t"<<getId()<<"\t   "<<name<<"\t";
        for (int i=0; i<5; i++)
            cout<<marks[i]<<" ";
        cout<<"|"<<endl;
    }
    else {
        cout<<"Id: "<<getId()<<"\nName: "<<name<<"\nMarks: ";
        for (int i=0; i<5; i++)
            cout<<marks[i]<<" ";
        cout<<endl;
    }
}

void displayAll (int c) {
    cout<<"\nSTUDENT DETAILS:\n\n";
    cout<<"+-------------------------------------------------------+\n";
    cout<<"|\tID\t|     \tNAME\t    |      MARKS       |\n";
    cout<<"+-------------------------------------------------------+\n";
    while(c--)
        s[c].showData(true);
    cout<<"+-------------------------------------------------------+\n";
}

int searchStudentById (int id, int c) {                 // returns index of successful
search
    for (int i=0; i<c; i++) {
        if (s[i].getId() == id)
            return i;
    }
    return -1;
    // cout<<"\nStudent with the given id: "<<id<<" not found in record!\n";
}

int searchStudentByName (string name, int c) {
    for (int i=0; i<c; i++) {
        if (s[i].name == name)
            return i;
    }
    return -1;
}

void updateStudent (int id, int c) {
    int idx = searchStudentById(id, c);
    string name;
    int marks[5];
    if (idx != -1) {
        cout<<"\nEnter new name of student (press 1 if update not required): ";
        cin>>name;
```

```
        if (name != "1")
            s[idx].name = name;
        cout<<"\nEnter new marks of student (press 1 if update not required): ";
        cin>>marks[0];
        if (marks[0] != 1) {
            s[idx].marks[0] = marks[0];
            for (int i=1; i<5; i++) {
                cin>>marks[i];
                s[idx].marks[i] = marks[i];
            }
        }
    }
}

int deleteStudentById(int id, int c) {
    int idx = searchStudentById(id, c);
    for(int i=idx; i<c-1; i++)
        s[i] = s[i+1];
    return c-1;
}

void sortById (int c) {
    int i,j,k;
    for (i=0; i<c-1; i++) {
        for (j=k=i; j<c; j++) {
            if (s[j].getId() > s[k].getId())
                k = j;
        }
        swap(s[i], s[k]);
    }
}

void sortByName (int c) {
    int i,j,k;
    for (i=0; i<c-1; i++) {
        for (j=k=i; j<c; j++) {
            if (s[j].name > s[k].name)
                k = j;
        }
        swap(s[i], s[k]);
    }
}


int main() {
    cout<<"\nFAIZAN CHOUDHARY\n20BCS021\n";

    int count = 0;
    int c, id, idx;
    string name;

    while (true) {
        cout<<"\nMENU:\n";
```

```
        cout<<"1. Add Student\n2. Update Student\n3. Delete Student by Id\n4. Display all
students\n5. Search by Id\n6. Search by name\n7. Sort by Id\n8. Sort by name\n9. Exit\n";
        cin>>c;
        switch(c) {
            case 1: if (count == 100) {
                        cout<<"\nMaximum data entry limit reached!\n";
                        break;
                    }
                    s[count++].getData();
                    //s[count].showData(false);
                    cout<<"\nData entry saved!\n";
                    break;
            case 2: cout<<"\nEnter id: ";
                    cin>>id;
                    updateStudent(id, count);
                    cout<<"\nData entry updated!\n";
                    break;
            case 3: cout<<"\nEnter id: ";
                    cin>>id;
                    count = deleteStudentById(id, count);
                    cout<<"\nData entry deleted!\n";
                    break;
            case 4: displayAll(count);
                    break;
            case 5: cout<<"\nEnter id: ";
                    cin>>id;
                    if (searchStudentById(id, count) == -1)
                        cout<<"\nStudent with the given id not found in record!\n";
                    else {
                        idx = searchStudentById(id, count);
                        cout<<"\nStudent found with the Id!\n";
                        s[idx].showData(false);
                    }
                    break;
            case 6: cout<<"\nEnter name: ";
                    cin.get();
                    getline(cin, name);
                    if (searchStudentByName(name, count) == -1)
                        cout<<"\nStudent with the given name not found in record!\n";
                    else {
                        idx = searchStudentByName(name, count);
                        cout<<"\nStudent found with the name!\n";
                        s[idx].showData(false);
                    }
                    break;
            case 7: sortById(count);
                    cout<<"\nAfter sorting:\n";
                    displayAll(count);
                    break;
            case 8: sortByName(count);
                    cout<<"\nAfter sorting:\n";
                    displayAll(count);
                    break;
            case 9: exit(0);
```

```
            default: cout<<"\nWrong Choice!\nEnter again!\n";
        }
    }
    return 0;
}
```

OUTPUT:

```
FAIZAN CHOUDHARY
20BCS021

MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
1

Enter details for student:

Enter student id: 23

Enter name of the student: Faizan Choudhary

Enter marks in 5 subjects (out of 100): 98 91 93 89 92

Data entry saved!
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
4

STUDENT DETAILS:

+-------------------------------------------------+
|       ID       |       NAME       |      MARKS      |
+-------------------------------------------------+
|       23       |  Faizan Choudhary  |  98 91 93 89 92 |
+-------------------------------------------------+
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
1

Enter details for student:

Enter student id: 29

Enter name of the student: Mohammad Adil

Enter marks in 5 subjects (out of 100): 93 91 90 94 92

Data entry saved!
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
4

STUDENT DETAILS:

+-------------------------------------------------+
|       ID       |       NAME       |      MARKS      |
+-------------------------------------------------+
|       29       |  Mohammad Adil   |  93 91 90 94 92 |
|       23       |  Faizan Choudhary  |  98 91 93 89 92 |
+-------------------------------------------------+
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
1

Enter details for student:

Enter student id: 69

Enter name of the student: Mohammad Kashif

Enter marks in 5 subjects (out of 100): 99 94 91 94 92

Data entry saved!
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
2

Enter id: 69

Enter new name of student (press 1 if update not required): 1

Enter new marks of student (press 1 if update not required): 99 90 93 94 92

Data entry updated!
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
5

Enter id: 23

Student found with the Id!
Id: 23
Name: Faizan Choudhary
Marks: 98 91 93 89 92
```

```
Enter name: Mohammad Kashif

Student found with the name!
Id: 69
Name: Mohammad Kashif
Marks: 99 90 93 94 92
```

```
After sorting:

STUDENT DETAILS:

+-----------------------------------------------------+
|       ID      |       NAME        |      MARKS      |
+-----------------------------------------------------+
|       23      |   Faizan Choudhary    98 91 93 89 92 |
|       29      |   Mohammad Adil       93 91 90 94 92 |
|       69      |   Mohammad Kashif     99 90 93 94 92 |
+-----------------------------------------------------+
```

```
MENU:
1. Add Student
2. Update Student
3. Delete Student by Id
4. Display all students
5. Search by Id
6. Search by name
7. Sort by Id
8. Sort by name
9. Exit
4

STUDENT DETAILS:

+-----------------------------------------------------+
|       ID      |       NAME        |      MARKS      |
+-----------------------------------------------------+
|       69      |   Mohammad Kashif     99 94 91 94 92 |
|       29      |   Mohammad Adil       93 91 90 94 92 |
|       23      |   Faizan Choudhary    98 91 93 89 92 |
+-----------------------------------------------------+
```

```
STUDENT DETAILS:

+-----------------------------------------------------+
|       ID      |       NAME        |      MARKS      |
+-----------------------------------------------------+
|       69      |   Mohammad Kashif     99 90 93 94 92 |
|       29      |   Mohammad Adil       93 91 90 94 92 |
|       23      |   Faizan Choudhary    98 91 93 89 92 |
+-----------------------------------------------------+
```

```
Enter id: 69

Data entry deleted!
```

```
STUDENT DETAILS:

+-----------------------------------------------------+
|       ID      |       NAME        |      MARKS      |
+-----------------------------------------------------+
|       23      |   Faizan Choudhary    98 91 93 89 92 |
|       29      |   Mohammad Adil       93 91 90 94 92 |
+-----------------------------------------------------+
```

FAIZAN CHOUDHARY

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CODE: (code pasted in this format for readability)

```
#include <bits/stdc++.h>
using namespace std;

class Person {
    string name;
    string address;
public:
    void setData(string n, string a) {
        name = n;
        address = a;
    }
    void display() {
        cout<<"\nName: "<<name<<endl<<"Address: "<<address<<endl;
    }
};

class Student: public Person {
    double per;
public:
    void setData(string name, string address, double per){
        Person::setData(name, address);
        this->per = per;
    }
    void displayStudent() {
        Person::display();
        cout<<"Percentage: "<<per<<endl;
    }
};

class Staff: public Person {
    double sal;
public:
    void setData(string name, string address, double sal){
        Person::setData(name, address);
        this->sal = sal;
    }
    void displayStaff() {
        Person::display();
        cout<<"Salary: "<<sal<<endl;
    }
};

int main() {
    string name, address;
```

```
    double sal, per;

    Student *std = new Student[3];
    Staff *stf = new Staff[3];

    for (int i=0; i<3; i++) {
        cout<<"\nEnter the details of student "<<i+1<<"-"<<endl;
        cout<<"Enter name: ";
        getline(cin, name);
        cout<<"\nEnter address: ";
        getline(cin, address);
        cout<<"\nEnter percentage: ";
        cin>>per;
        cin.ignore();
        std[i].setData(name, address, per);
    }

    for (int i=0; i<3; i++) {
        cout<<"\nEnter the details of staff "<<i+1<<"-"<<endl;
        cout<<"Enter name: ";
        getline(cin, name);
        cout<<"\nEnter address: ";
        getline(cin, address);
        cout<<"\nEnter salary: ";
        cin>>sal;
        cin.ignore();
        stf[i].setData(name, address, sal);
    }

    cout<<"\nDetails of students-"<<endl;
    for (int i=0; i<3; i++) {
        std[i].displayStudent();
    }

    cout<<"\nDetails of staff-"<<endl;
    for (int i=0; i<3; i++) {
        stf[i].displayStaff();
    }

    return 0;
}
```

OUTPUT:

```
Enter the details of student 1-
Enter name: Faizan Choudhary

Enter address: Batla House

Enter percentage: 96

Enter the details of student 2-
Enter name: Soban farooq

Enter address: Bareilly

Enter percentage: 92

Enter the details of student 3-
Enter name: Mohd Adil

Enter address: Punjab

Enter percentage: 93

Enter the details of staff 1-
Enter name: Fardeen Hussain

Enter address: Lajpat Nagar

Enter salary: 15000

Enter the details of staff 2-
Enter name: Fazeel Rizvi

Enter address: Rajasthan

Enter salary: 34000

Enter the details of staff 3-
Enter name: Mohd Kashif

Enter address: Sitapur

Enter salary: 85000
```

```
Details of students-

Name: Faizan Choudhary
Address: Batla House
Percentage: 96

Name: Soban farooq
Address: Bareilly
Percentage: 92

Name: Mohd Adil
Address: Punjab
Percentage: 93


Details of staff-

Name: Fardeen Hussain
Address: Lajpat Nagar
Salary: 15000

Name: Fazeel Rizvi
Address: Rajasthan
Salary: 34000

Name: Mohd Kashif
Address: Sitapur
Salary: 85000
```

FAIZAN CHOUDHARY

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CODE: (code pasted in this format for readability)

```
#include <bits/stdc++.h>
using namespace std;

class Time{
private:
    int hour;
    int minute;

public:
    Time(){};
    Time(int hours, int minutes)
    {
        this->hour = hours;
        this->minute = minutes;
    }
    Time operator+(Time t)
    {
        Time temp;
        temp.hour = hour + t.hour;
        temp.minute = minute + t.minute;
        if (temp.minute >= 60)
        {
            temp.hour++;
            temp.minute -= 60;
        }
        return temp;
    }
    friend void display(Time t);
};

void display(Time t){
    cout << t.hour << ":" << t.minute << endl;
}
int main()
{
    Time t1(1, 45), t2(3, 30), t3;
    cout << "Time t1: ";
    display(t1);
    cout << "Time t2: ";
    display(t2);
    t3 = t1 + t2;
    cout << "Time t3 = t1 + t2: ";
    display(t3);
    return 0;
```

```
}
```

OUTPUT:

```
Time t1: 1:45
Time t2: 3:30
Time t3 = t1 + t2: 5:15
```

FAIZAN CHOUDHARY

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CODE: (code pasted in this format for readability)

```
#include <bits/stdc++.h>
using namespace std;

class Media
{
protected:
    string title;
    string publication;

public:
    Media(){};
    Media(string title, string publication)
    {
        this->title = title;
        this->publication = publication;
    }
    virtual void read(){
        cout << "Reading Media" << endl;
    }
    virtual void show(){
        cout << "Watching Media" << endl;
    }
    virtual ~Media(){
        cout << "Media Deleted" << endl;
    }
};

class Book : public Media
{
protected:
    int pages;

public:
    Book(string title, string publication, int pages) : Media(title, publication){
        this->pages = pages;
    }
    void read(){
        cout << "Reading " << title << " published by " << publication << endl;
    }
    ~Book(){
        cout << "Book deleted" << endl;
    }
};
class Tape : public Media
```

```
{
protected:
    int playing_time;

public:
    Tape(string title, string publication, int playing_time) : Media(title, publication){
        this->playing_time = playing_time;
    }
    void show()
    {
        cout << " Playing " << title << " published by " << publication << endl;
    }
    ~Tape()
    {
        cout << "Tape Deleted" << endl;
    }
};

int main()
{
    Media *media;
    string title, publication;
    int pages, time;
    cout << "Enter title of book: ";
    getline(cin, title);
    // getchar();
    cout << "Enter publication: ";
    getline(cin, publication);
    cout << "Enter no. of pages: ";
    cin >> pages;
    media = new Book(title, publication, pages);
    media->read();
    getchar();
    cout << "\nEnter title of tape: ";
    getline(cin, title);
    cout << "Enter publication: ";
    getline(cin, publication);
    cout << "Enter watch time in minutes: ";
    cin >> time;
    media = new Tape(title, publication, time);
    media->show();
    return 0;
}
```

OUTPUT:

```
Enter title of book: It ends with us
Enter publication: Collen Hoover
Enter no. of pages: 365
Reading It ends with us published by Collen Hoover

Enter title of tape: The Boy with Love
Enter publication: Penguin
Enter watch time in minutes: 45
Playing The Boy with Love published by Penguin
```

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CODE: (code pasted in this format for readability)

```
#include<iostream>
using namespace std;

struct account{
    int accNO;
    string name;
};

class AccBook{
    int noAcc;
    account a[10];

    public :
        AccBook(int n){
            noAcc=n;
        }

        void setData(){
            for(int i=0; i<noAcc; i++){
                cout<<"\nEnter details of Record "<<i+1<<" : "<<endl;
                cout<<"Enter AccNO : ";
                cin>>a[i].accNO;

                cout<<"Enter Account Holder name : ";
                // cin>>a[i].name;
                getchar();
                getline(cin,a[i].name);
            }
        }

        void printBook(){

            cout<<"No of Accounts : "<<noAcc<<endl;

            for(int i=0; i<noAcc; i++){
                cout<<"Account Holder No : "<<a[i].accNO<<endl;
                cout<<"Account Holder Name : "<<a[i].name<<endl;
            }
        }

        string operator[] (int n){
            for(int i=0; i<noAcc; i++){
                if(a[i].accNO==n)
                    return a[i].name;
```

```
            }

            return "";
        }
};

int main()
{
    cout<<"Enter no of records : ";
    int n; cin>>n;
    AccBook o(n);
    o.setData();
    o.printBook();

    int x;
    cout<<"\nEnter the accNo of the person you want to find name of : ";
    cin>>x;
    cout<<o[x];

    return 0;
}
```

OUTPUT:

```
Enter no of records : 3

Enter details of Record 1 :
Enter AccNO : 6969
Enter Account Holder name : Faizan Choudhary

Enter details of Record 2 :
Enter AccNO : 1241
Enter Account Holder name : Soban Farooq

Enter details of Record 3 :
Enter AccNO : 9217
Enter Account Holder name : Mohd Kashif

No of Accounts : 3
Account Holder No : 6969
Account Holder Name : Faizan Choudhary
Account Holder No : 1241
Account Holder Name : Soban Farooq
Account Holder No : 9217
Account Holder Name : Mohd Kashif
```

FAIZAN CHOUDHARY

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CODE: (code pasted in this format for readability)

```
#include<iostream>
using namespace std;

class complex{
    float real=0;
    float img=0;

    public:
        void form(int a, int b){
            real=a;
            img=b;
        }

        complex operator+(complex c){
            complex f;
            f.real = real + c.real;
            f.img = img + c.img;
            return f;
        }

        complex operator-(complex c){
            complex f;
            f.real = real - c.real;
            f.img = img - c.img;
            return f;
        }

        complex operator*(complex c){
            complex f;
            f.real = real*c.real - img*c.img;
            f.img = real*c.img + img*c.real;
            return f;
        }

        complex operator/(complex c){
            complex f;
            float deno = real*c.real - img*c.img;
            f.real = (real*c.real + img*c.img)/deno;
            f.img = (img*c.real - real*c.img)/deno;
            return f;
        }

        friend ostream &operator<<(ostream &o, complex c);
};
```

```
ostream &operator<<(ostream &o, complex c){
    o<<c.real<<" + "<<c.img<<"i\n";
    return o;
}

int main()
{
    complex c1,c2;
    c1.form(6,1);
    c2.form(8,0);

    cout<<c1;
    cout<<c2;

    cout<<c2+c1;
    cout<<c2-c1;
    cout<<c1*c2;
    cout<<c2/c1;
    return 0;
}
```

OUTPUT:

```
6 + 1i
8 + 0i
14 + 1i
2 + -1i
48 + 8i
1 + -0.166667i
```

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CODE: (code pasted in this format for readability)

```
#include<iostream>
using namespace std;

class Dollar;

class Rupee{
    private :
        float rup;

    public :
        Rupee(){
            rup=0.0;
        }

        Rupee(float r){
            rup=r;
        }

        Rupee(Dollar d);
        void setRupee(){
            cout<<"Enter Amount in Indian Rupees : ";
            cin>>rup;
        }

        float getRupee(){ return rup; }

        void show(){ cout<<"Indian Rupee : "<<getRupee(); }
};

class Dollar{
    private :
        float dol;

    public :
        Dollar(){
            dol=0.0;
        }

        Dollar(float d){
            dol=d;
        }

        Dollar(Rupee r){
            dol = r.getRupee() * 0.0121;
```

```
        }

        void setDollar(){
            cout<<"Enter Amount in US Dollars : ";
            cin>>dol;
        }

        float getDollar(){ return dol; }

        void show(){ cout<<"US Dollar : "<<getDollar(); }
};

Rupee::Rupee(Dollar d){
    rup = d.getDollar() * 82.38;
}
int main(){

    cout<<"\nCONVERSION ROUTINE IN DESTINATION CLASS : USING CONSTRUCTOR\n\n";

    cout<<"US Dollar to Indian Rupees : \n";
    Dollar d1;
    Rupee r1;

    d1.setDollar();
    r1 = d1;
    r1.show();


    cout<<"\n\nIndian Rupees to US Dollars : \n";
    Dollar d2;
    Rupee r2;

    r2.setRupee();
    d2 = r2;
    d2.show();

    return 0;
}
```

OUTPUT:

```
CONVERSION ROUTINE IN DESTINATION CLASS : USING CONSTRUCTOR

US Dollar to Indian Rupees :
Enter Amount in US Dollars : 30
Indian Rupee : 2471.4

Indian Rupees to US Dollars :
Enter Amount in Indian Rupees : 1200
US Dollar : 14.52
```

FAIZAN CHOUDHARY

20BCS021

OOPs LAB

CODE: (code pasted in this format for readability)

```
#include <iostream>
using namespace std;

class base
{
public:
    base()
    {
        cout << "Base class constructor" << endl;
    }
    // Pure Virtual Function
    virtual void display() = 0;
    // virtual destructor
    virtual ~base()
    {
        cout << "Destructor of base class" << endl;
    }
};

class derived : public base
{
public:
    derived()
    {
        cout << "Derived class constructor" << endl;
    }
    void display()
    {
        cout << "Derived class display function" << endl;
    }

    // destructor
    ~derived()
    {
        cout << "Derived class destructor" << endl;
    }
};

// overloading through friend function
class Rect{
    int a, b;

public:
    Rect(int x, int y){
```

```
        a = x;
        b = y;
    }
    friend int operator+(Rect, Rect);
    friend ostream &operator<<(ostream &, Rect);
};

int operator+(Rect r1, Rect r2){
    return r1.a * r1.b + r2.a * r2.b;
}

ostream &operator<<(ostream &out, Rect c){
    out << "L = "<<c.a << ", B = " << c.b << endl;
    return out;
}
int main()
{
    base *b;
    derived d;
    b = &d;
    b->display();
    Rect r1(2, 3), r2(4, 5);
    cout<<r1;
    cout<<r2;
    cout << "Total Area : " <<r1 + r2 <<endl;
    return 0;
}
```

OUTPUT:

```
Base class constructor
Derived class constructor
Derived class display function
L = 2, B = 3
L = 4, B = 5
Total Area : 26
Derived class destructor
Destructor of base class
```

FAIZAN CHOUDHARY

20BCS021

OOPs LAB

CODE: (code pasted in this format for readability)

```
abstract class Shape {
    int p1;
    int p2;

    abstract void printArea();
}

class Triangle extends Shape {
    Triangle(int height, int base) {
        p1 = height;
        p2 = base;
    }

    void printArea() {
        System.out.println("Area of Triangle is: " + (0.5 * p1 * p2));
    }
}

class Rectangle extends Shape {
    Rectangle(int length, int breadth) {
        p1 = length;
        p2 = breadth;
    }

    void printArea() {
        System.out.println("Area of Rectangle is: " + (p1 * p2));
    }
}

class Circle extends Shape {
    Circle(int radius) {
        p1 = radius;
    }

    void printArea() {
        System.out.println("Area of Circle is: " + (3.14 * p1 * p1));
    }
}

public class program9 {

    public static void main(String[] args) {
        Triangle t = new Triangle(15, 10);

        Rectangle r = new Rectangle(25, 10);
```

```
        Circle c = new Circle(5);
        t.printArea();
        r.printArea();
        c.printArea();
    }
}
```

OUTPUT:

```
Area of Triangle is: 75.0
Area of Rectangle is: 250
Area of Circle is: 78.5
```

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20BCS021

OOPs LAB

CODE: (code pasted in this format for readability)

```
import java.util.Random;

class Square extends Thread {
    int x;

    Square(int n) {
        x = n;
    }

    public void run() {
        int sqr = x * x;
        System.out.println("Square of " + x + " = " + sqr);
    }
}

class Cube extends Thread {
    int x;

    Cube(int n) {
        x = n;
    }

    public void run() {
        int cub = x * x * x;
        System.out.println("Cube of " + x + " = " + cub);
    }

}

class Number extends Thread {

    public void run() {
        Random random = new Random();

        while (true) {
            int randomInteger = random.nextInt(100);
            System.out.println("\nRandom Integer generated : " + randomInteger);

            if (randomInteger % 2 == 0) {
                Square square = new Square(randomInteger);
                square.start();
            } else {
                Cube cube = new Cube(randomInteger);
                cube.start();
```

```
            }

            try {
                Thread.sleep(1000);
            } catch (InterruptedException ex) {
                System.out.println(ex);
            }
        }
    }
}

public class program10 {
    public static void main(String args[]) {
        Number n = new Number();
        n.start();
    }
}
```

OUTPUT:

```
Random Integer generated : 34
Square of 34 = 1156

Random Integer generated : 75
Cube of 75 = 421875

Random Integer generated : 96
Square of 96 = 9216

Random Integer generated : 9
Cube of 9 = 729

Random Integer generated : 75
Cube of 75 = 421875

Random Integer generated : 52
Square of 52 = 2704

Random Integer generated : 64
Square of 64 = 4096

Random Integer generated : 71
Cube of 71 = 357911

Random Integer generated : 13
Cube of 13 = 2197

Random Integer generated : 93
Cube of 93 = 804357
```